

UNIVERSITI TUNKU ABDUL RAHMAN
Faculty of Information and Communication Technology



UCCD3223 Mobile Applications Development
(February 2026 Trimester)

Group Assignment 1

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Marking scheme	Marks	Remarks
Creativity/novelty of concept	X 4	
Completeness of idea	X 3	
Overall design	X 2	
Neat documentation layout	X 1	
TOTAL	/100	

1.Introduction

1.1 Project Background

CharityLink is a mobile application developed under the UCCD3223 Mobile Application Development course. This application is a dedicated social platform designed to facilitate the free flow of unwanted household items and the donation of food, daily necessities, and more to charities between local community donors and recipients. CharityLink simplifies cumbersome logistics processes, prioritizing social posting and real-time chat features to create a relaxed and convenient communication environment, thereby encouraging people to develop habits of donation, reuse, and recycling.

1.2 Core Purpose and Motivation

Everywhere in the world is turning into a mobile connected place, CharityLink makes use of the omnipresence of smartphone devices to address the issue of domestic waste in the real world. The application is aimed at niche markets, including students at a university, residents of a particular neighborhood, charity organization who tend to have used such items like textbooks, kitchenware, or clothes that are no longer needed but they are reluctant to dispose of them. The app will offer a direct means of communication and thus the second life of these items will be achieved hence a purposeful social impact.

1.3 Alignment with Sustainable Development Goals (SDG)

This project is built to contribute to the following **Sustainable Development Goals (SDG)**:

- **Goal 12:** Responsible Consumption and Production: With the ability to redistribute used goods, the app might help in decreasing the level of waste in landfills.
- **Goal 11:** Sustainable Cities and Communities: The platform enhances the collaboration of the local authority and fortifies connections within a community by sharing via P2P.
- **Goal 4:** Quality Education: The free sharing of educational resources such as books and stationeries in specific student communities with the help of the app.

- **Goal 1: No Poverty:** This will enable the poor or less affluent members of society to afford essential commodities hence make them less susceptible by identifying with financial atrocities.

1.4 Technical Scope

To ensure high performance and a native experience, the application is developed using **Android Studio**. It satisfies all minimum assessment criteria, including:

- A **custom-designed icon** representing the brand identity.
- Integration with **External Web Services** for real-time social feeds and instant messaging.
- The use of **Cloud Storage (Database)** to manage user profiles and post history tracking on the app.

2. Problem Statement

The Emotional & Physical Burden of Clutter

Problem: Most people have useful things (textbooks, clothes, electronics) that are not in use to them but have value. The guilt of waste makes them not want to dispose of things in the trash, but the absence of an outlet to do so causes clutter at home.

Gap: The current platforms (such as Facebook Marketplace or Carousell) are usually too concentrated on making a money profit, so the process of listing a free item does not seem worth it.

The Logistical Friction Problem

Problem: Traditional charity organizations often require donors to travel to specific drop-off points during limited operating hours.

Gap: For users without transport, the cost of time and travel of donating a small item exceeds the perceived benefit. **CharityLink** solves this by focusing on local (neighborhood or state) exchanges.

The Trust & Security Gap in Peer-to-Peer Giving

Problem: Users might not be safe or secure to give things to strangers through anonymous boards (ex. Craigslist).

Gap: It does not have a community-first platform with user profiles, social feed, and ratings. This is what **CharityLink** fulfills by providing a socially-verified environment.

3.1 Proposed Solution

CharityLink is the name of our planned mobile App, which represents a dedicated non-monetary platform of charitable contributions based on items donation in Malaysia. In contrast to the traditional fundraising apps where the company mainly concentrates on monetary donations, CharityLink is only devoted to redistributing the necessary reusable items like clothes, textbooks, toys, school supplies, and household products.

The site is an online platform that links donors with verified charity organizations in a central and digitized space. By making the application focused on physical goods instead of money, the application will help households to be clean, minimize waste, and help them become responsible consumers.

3.1.1 Submission System of item donation.

Through the application, the user can post what they would like to donate including clothes, books, toys, school supplies, and other reusable household goods. The user is allowed to post pictures of the items, choose a category, give a little description, and specify the condition of items. After it has been submitted, the donation listing can be seen by registered charity organizations on the platform.

3.1.2 Charity Needs Posting System.

Charity organizations will be able to form and set up posts about the particular items that they need at the moment. The requests are arranged in a systematic and tabulated manner, enabling the user to navigate through the requests easily and find matching items to be donated.

3.1.3 Chat Communication System.

The application also has an inbuilt chat system which can be used to directly contact the donor and charity organizations. After a donor posts an item, the two parties are able to interact in the app to share more information like clarification of items or handover arrangements. The text messages are logged in the application and can be accessed to refer to.

3.1.4 User Profile and Donation Post History.

Every user has a profile page that shows a list of past donation postings made by him. The users are able to look at the details of items and history with the charities. The records of the donations are saved on the mobile device locally so that the data can be accessed and updated accordingly.

3.1.5 State-Based Filtering System

The site will have a filtering option, which will enable users to search and browse the list of donation based on the type of items and geographical location. This option enhances the convenience of users and provides the efficiency of local communities in the process of matching the items. This will allow donors and charity organizations to find relevant listings and avoid the situation whereby they must scroll through all the listings available. Moreover, the system enables users to sort listings by state so that donations would not be distributed beyond the area. Such a local strategy meets the aim of the platform of helping to make the process of item-based donation practical and accessible without long-range travel.

3.2 Contribution

3.2.1 Reducing Waste and Responsible Consumption.

Among the main advantages of CharityLink, it should be noted that the product encourages responsible consumption and waste minimization. The application motivates people to reuse the products rather than to accumulate them in landfills by offering a platform that serves as a dedicated location to all the reusable products.

3.2.2 Decreasing Emotional Load and Promoting Positive Alteration in Behavior.

The emotional weight of clutters and idle items is also a concern of CharityLink. The system that is easy and structured regarding giving items eliminates the psychological barrier to giving them and makes the process of giving a donation easier. This can inspire users to develop regular habits of donating to encourage a culture of responsible sharing and redistribution as opposed to needless hoarding.

3.2.3 Expanding Accessibility to Charitable Engagement.

The conventional charity models normally place the burden on the donors to travel to the physical drop off points and within restricted working times. This can make it discouraging especially to persons who have less time or transport. CharityLink makes it more accessible as it concentrates on coordination in a mobile application, which gives users more flexibility in arranging the donation of items. The platform makes the participation inclusive and open to a wider range of the population simply because it eliminates the necessity of any monetary contributions and focuses on the basic goods.

3.2.4 Investing Trust and Structured Giving.

The other important contribution is that there has been the creation of a transparent and well-organized environment of donation. CharityLink is an online platform that is unique in offering a redistribution of essential items in a specific digital environment where chat messages are recorded within an in-built chat platform. The risks of the online monetary fraud are eliminated since the platform does not handle financial transactions. This builds confidence between the charity organizations and donors and improves the credibility of the whole process of donation.

3.2.5 Improving the efficiency of the operations of Charities.

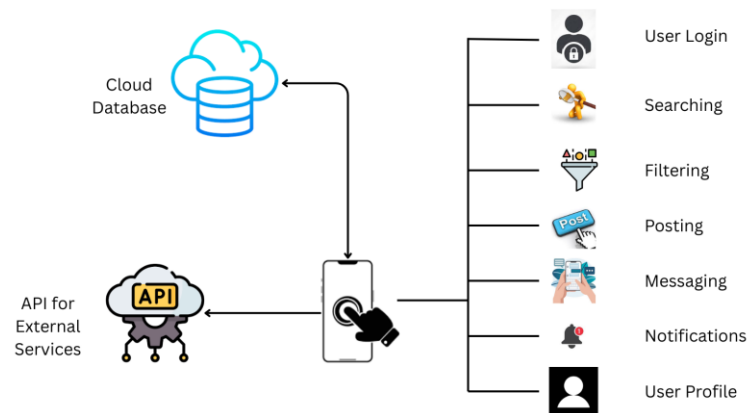
CharityLink will enhance the levels of efficiency in operations since charities will be able to leave certain needs regarding items. This model of demand ensures that the donations are in accordance with the real needs and will lessen the problems of overloading of the storage, wrong donations, or misplacing the resources. This means that the charity organizations will have less speculative

donations to attend to since they can now concentrate on giving the beneficiaries the needs and donations that are vital.

4.1 System Architecture

The application follows a client-server architecture to ensure data persistence and real-time updates.

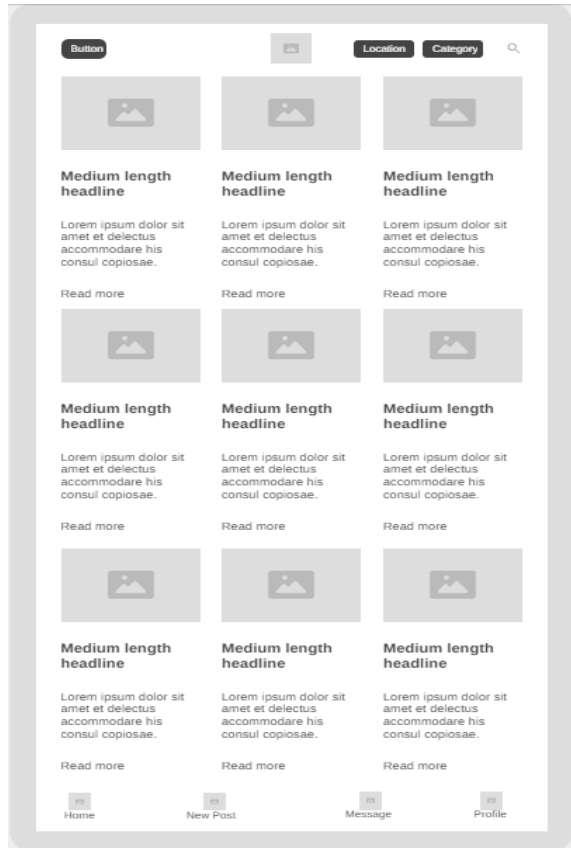
- **Mobile Client:** Developed using **Android Studio**.
- **Local Storage:** Employs **Room Database** to store user history and offline badges.
- **External Services:** Uses **Google Play Services Location API** for post suggestion services, **Google ML Kit** for image processing, **REST API for real-time data communication between the mobile app and cloud database**, and **World Time API** to retrieve accurate time data based on user location.
- **Backend:** Uses a cloud database like Firebase or others to store user profiles, community posts, comment and charity information with real-time synchronization.



4.2 Mock-up UI Design

The UI follows a clean, "Eco-Friendly" green theme to reflect the sustainability goals. Key screens include:

1. **Home Feed:** A list of recent community donations and success stories.



2. **Messaging Function:** A messaging screen where users can chat and receive or reply when interacting with other users.



3. **Posting Featured:** A screen where users allow to create a new post and share it with other users to view, like, and comment in real time.



4. **User profile:** A personalized page showing user profile detail and posts have been posted by user.

